LoRa

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Problem Domain

- LoRa(Long Range) has a low bandwidth capacity, which allows for long range, but small payloads. Cannot send large files effectively.

Proposed Solution

Proposed Solution

- Understand the **limitations** of **LoRa** in **multiple** different **network settings**.
- Create a extendable P2P network to transfer files effectively

Architecture

System Overview pt. 1

Physical Components:

- **ESP32 Micro Controller**: **Brains** and main component running software
- **SD Card Reader** - for storage of transmission objects and logging
- **GPS/GNSS** - Accurate logging position
- **LoRa Tx/Rx**: Transmission and reception devices

System Overview pt. 2

Tech Stack:

- **VsCode Arduino Maker Workshop**: Organize, compile and upload software
- **Copilot - ChatGPT - Claude** - Questions, MVP
- **C++** - Code design
- **Github** - VCS

Sprint 1

Goals

- Simple **LoRa P2P network**
- Understand LoRa's **Spreading Factor** and **Bandwidth**

Tasks

1. Understand how to make a **simple LoRa network** with **complete payload reception** and **transmission**
2. Push it to it's limits - **understand** the **limitations** of the devices (Bandwidth, range)
3. Create **evaluation matrix** for **combinations** of **Spreading factor and bandwidth**.

Sprint 2

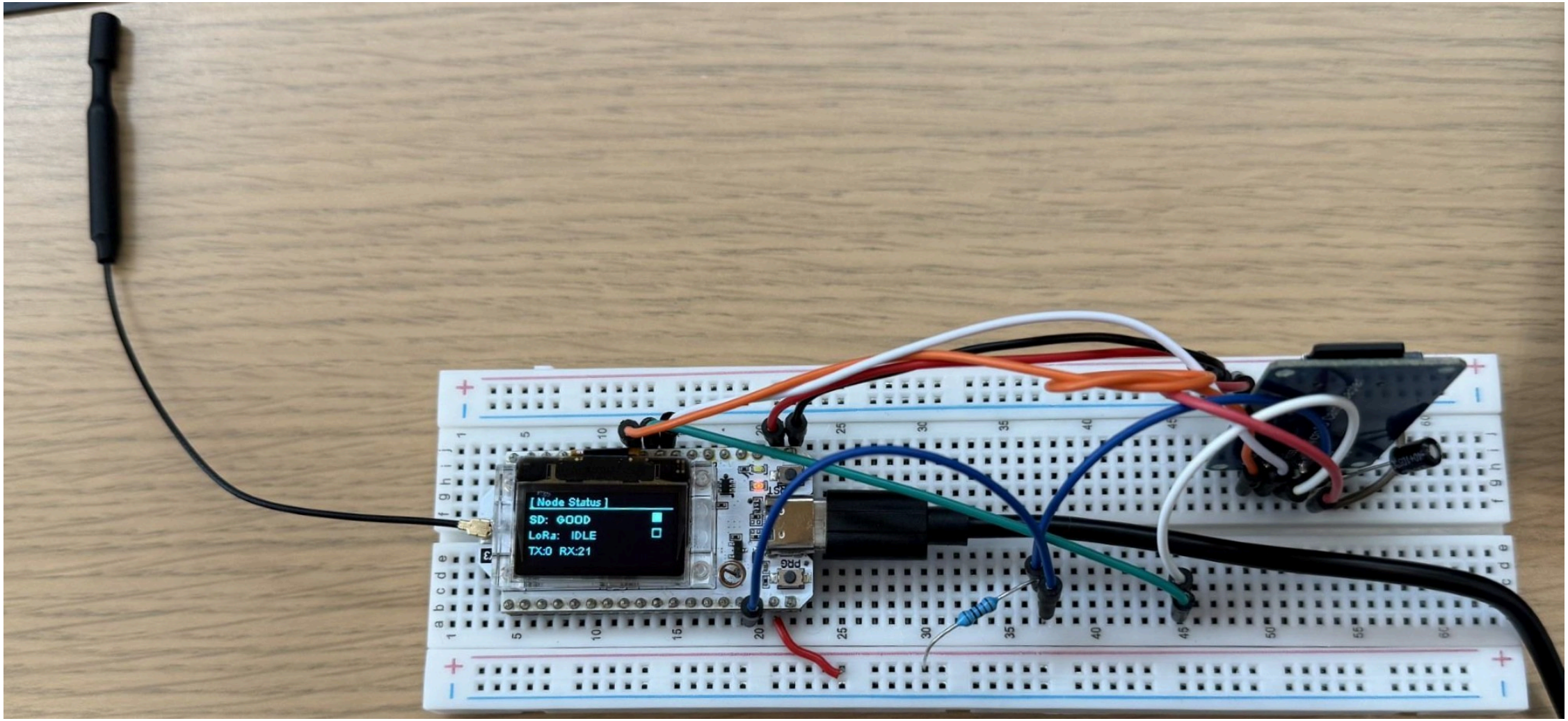
Goals

- Update to state machine for further testing
- Add GPS/GNSS
- Finish Paper

Tasks

1. Create a better more extenadable implementation of the device for further testing
2. Add GPS/GNSS for
3. Write and submit paper with results

Images -



Images 2 -

Monitor Mode Serial View Mode Text Port COM7 - Silicon Labs CP210x USB to UART Bridge (COM7) Baud rate 115200 Line ending CRLF

Stop Monitoring

```
[SD] LOG WRITE OK type=START status=ACK_OK sid=5803 seq=32 frag=-1 len=0 tx=41956 ack=42323 rtt=367ms RSSI=-44 SNR=12.8
[SD] CSV: 42525,41956,42323,367,0.000000,0.000000,-44,12.8,5803,32,-1,0,START,ACK_OK
[TX] AUDIO_DATA seq=33 len=128 sent
[RX] ACK OK for seq=33  RSSI=-45.0  SNR=12.2
[LOG] write csv row type=DATA status=ACK_OK tx=42761 ack=43298 rssi=-45 snr=12.2
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=33 frag=0 len=128 tx=42761 ack=43298 rtt=537ms RSSI=-45 SNR=12.2
[SD] CSV: 43500,42761,43298,537,0.000000,0.000000,-45,12.2,5803,33,0,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=34 len=128 sent
[RX] ACK OK for seq=34  RSSI=-44.0  SNR=12.8
[LOG] write csv row type=DATA status=ACK_OK tx=43586 ack=44116 rssi=-44 snr=12.8
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=34 frag=1 len=128 tx=43586 ack=44116 rtt=530ms RSSI=-44 SNR=12.8
[SD] CSV: 44318,43586,44116,530,0.000000,0.000000,-44,12.8,5803,34,1,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=35 len=128 sent
[RX] ACK OK for seq=35  RSSI=-44.0  SNR=12.2
[LOG] write csv row type=DATA status=ACK_OK tx=44410 ack=44942 rssi=-44 snr=12.2
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=35 frag=2 len=128 tx=44410 ack=44942 rtt=532ms RSSI=-44 SNR=12.2
[SD] CSV: 45144,44410,44942,532,0.000000,0.000000,-44,12.2,5803,35,2,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=36 len=128 sent
[RX] ACK OK for seq=36  RSSI=-45.0  SNR=12.5
[LOG] write csv row type=DATA status=ACK_OK tx=45230 ack=45767 rssi=-45 snr=12.5
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=36 frag=3 len=128 tx=45230 ack=45767 rtt=537ms RSSI=-45 SNR=12.5
[SD] CSV: 45968,45230,45767,537,0.000000,0.000000,-45,12.5,5803,36,3,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=37 len=128 sent
[RX] ACK OK for seq=37  RSSI=-44.0  SNR=12.2
[LOG] write csv row type=DATA status=ACK_OK tx=46054 ack=46584 rssi=-44 snr=12.2
[SD] LOG WRITE OK type=DATA status=ACK_OK sid=5803 seq=37 frag=4 len=128 tx=46054 ack=46584 rtt=530ms RSSI=-44 SNR=12.2
[SD] CSV: 46786,46054,46584,530,0.000000,0.000000,-44,12.2,5803,37,4,128,DATA,ACK_OK
[TX] AUDIO_DATA seq=38 len=128 sent
[RX] ACK OK for seq=38  RSSI=-44.0  SNR=11.8
```

Learning with AI

AI Integration 1

C++ management

1. C++ is a highly optimizable language perfect for embedded system's programming. It can be useful but also tricky(especially to native Java users) with it's pointers, Structs, STD lib, and memory management just to name a few.

AI Integration 2

Digital Communication

2. There a couple over-arching methods of digital communication. Serial and Parallel. Serial is the most common is this day and age due to the speed of cable comms. It has a bunch of protocols that fall under it such as SPI, I2C, UART, and USB to name a few. Wireless is predominantly in serial but is more defined at the transport layer with things lke Bluetooth, WiFi, LoRa, 433 Mhz to name a few.

How I used AI for coding

- Think about AGILE structure and being a mnager. AI is my subordinate
  - Create features and requirements and self document all. Send to AI, confirm all progress makes sense
  - Rinse and repeat
  - When in a good place, have AI create an integration plan and review, rinse and repeat **WITH SELF REVIEW**
  - Then have AI follow the plan and test along the way

Paper

link

Future

- Bit off more than I can chew with mesh network implementation
  - I have the framework and extension ready to continue but other work got in the way
- Continue testing and write another paper using mesh and

Questions?

Thank you for your attention!